National Curriculum of Pakistan 2022-2023

Assessment Framework Physics Grade-XII

Details of Content Areas/ SLOs

NCP SLOs Description	Form of Assessment	Cognitive domain	Remarks	Number of Periods Required (1 period=40 minutes)
Domain: B Mechanics				
[SLO: P-12-B-01] Define and calculate gravitational field strength [This will include more challenging problems than in Grade 9. It will involve use of $g = GM/r^2$].	Summative for theory	Knowledge + Understanding + Application	Question(s) will be asked in final examination	07 periods
[SLO: P-12-B-02] Analyze gravitational fields by means of field lines. [This includes knowing that for a point outside a uniform sphere, the mass of the sphere may be considered to be a point mass at its center].	Summative for theory	Application	Question(s) will be asked in final examination.	
[SLO: P-12-B-03] Apply Newton's law of gravitation to solve problems [$F=Gm_1m_2/r^2$ for the force between two-point masses to solve problems].	Summative for theory	Application	Question(s) will be asked in final examination.	
[SLO: P-12-B-04] Analyze circular orbits in gravitational fields [By relating the gravitational force to the centripetal acceleration, it causes].	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-B-05] Analyze the motion of geostationary satellites [This includes knowing that geostationary orbit remains at the same point above the Earth's surface, with an orbital period of 24 hours, orbiting from west to east, directly above the Equator].	Summative for theory	Application	Question(s) will be asked in final examination.	05 periods
[SLO: P-12-B-06] Derive the equation for gravitational field strength [From Newton's law of gravitation and the definition of gravitational field, the equation $g = GM/r^2$ for the gravitational field strength due to a point mass].	Summative for theory	Application	Question(s) will be asked in final examination.	
[SLO: P-12-B-07] Analyze why g is approximately constant for small changes in height near the Earth's surface.	Summative for theory and PBA	Understanding + Application	Question(s) will be asked in final theory examination and PBA.	

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[SLO: P-12-B-08] Define and calculate gravitational potential [Use $\varphi = -\frac{GM}{r}$ for the gravitational potential in the field due to a point mass]. [At a point as the work done per unit mass in bringing a small test mass from infinity to the point].	Summative for theory	Knowledge + Understanding + Application	Question(s) will be asked in final examination.	10 periods
[SLO: P-12-B-09] Justify how the concept of gravitational potential leads to the gravitational potential energy of two-point masses [Use of E_p =-G Mm/r in problems is expected].	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
Domain: C Heat and Thermodynamics				
[SLO: P-12-C-01] Explain how molecular movement causes the pressure exerted by gas.	Summative for theory	Understanding	Question(s) will be asked in final examination	
[SLO: P-12-C-02] Derive and use the relationship pV =Nm/3 $<c^2>$ [where $<c^2>$ is the mean-square speed (a simple model considering one-dimensional collisions and then extending to three dimensions. Using $1/3<c^2>=<c_x^2>$ is sufficient)].	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-C-03] Calculate the root-mean-square speed of an ideal gas.	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-C-04] Derive and use the formula for the average translational kinetic energy of a gas.	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-C-05] Illustrate that the model of ideal gasses is used a base from which the field of statistical mechanics emerged [and has helped explain the behavior of 'non-ideal' gasses through modifications to the model e.g. the behavior of stars].	Formative	Knowledge + Understanding	Question(s) will not be asked in final examination.	05 periods

NCP SLOs Description	Form of Assessment	Cognitive domain	Remarks	Number of Periods Required (1 period=40 minutes)
[SLO: P-12-C-06] State that under extreme physical conditions, atoms can break down into sub-atomic particle that can form unusual states of matter [Such as degenerate matter. Usually made of any one kind of subatomic particle such as neutron degenerate matter in neutron stars under strong gravity and heat) and Bose-Einstein condensates (Created when certain materials are taken to very low temperatures and then exhibit remarkable properties like superconductivity and super fluidity)].	Formative	Knowledge + Understanding	Question(s) will not be asked in final examination.	
Domain: D Waves				
Simple Harmonic Motion:				
[SLO: P-12-D-01] Describe simple examples of free oscillations.	Summative for theory	Knowledge + Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-D-02] Use the terms displacement, amplitude, period, frequency, angular frequency and phase difference in the context of oscillations.	Summative for theory	Knowledge + Understanding	Question(s) will be asked in final examination.	05 periods
[SLO: P-12-D-03] Express the period of simple harmonic motion in terms of both frequency and angular frequency.	Summative for theory and PBA	Understanding	Question(s) will be asked in final theory examination and PBA.	
[SLO: P-12-D-04] Explain that simple harmonic motion occurs when acceleration is proportional to displacement from fixed point and in the opposite direction.	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-D-05] Use $a = -\omega^2 x$ to solve problems.	Summative for theory	Application	Question(s) will be asked in final examination.	15 periods
[SLO: P-12-D-06] Use the equations $v = v_o \cos(\omega t)$ and $v = \pm\omega\sqrt{x_o^2 - x^2}$ to solve problems.	Summative for theory	Application	Question(s) will be asked in final examination.	
[SLO: P-12-D-07] Analyze graphical representations of the variations of displacement, velocity and acceleration for simple harmonic motion.	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-D-08] Analyze the interchange between kinetic and potential energy during simple harmonic motion.	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	

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[SLO: P-12-D-09] Apply $\frac{1}{2}m\omega^2x_0^2$ for the total energy of system undergoing simple harmonic motion.	Summative for theory	Application	Question(s) will be asked in final examination.	
[SLO: P-12-D-10] Describe that a resistive force acting on an oscillating system causes damping.	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-D-11] Use the terms light, critical and heavy damping.	Summative for theory	Knowledge+ Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-D-12] Sketch displacement-time graphs: to illustrate light, critical and heavy damping.	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-D-13] State that resonance involves a maximum amplitude of oscillations and that this occurs when an oscillating system is forced to oscillate at its natural frequency.	Summative for theory	Knowledge+ Understanding	Question(s) will be asked in final examination.	32 periods
[SLO: P-12-D-14] Describe practical examples of free and forced oscillations.	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-D-15] Describe practical examples of damped oscillations [with particular reference to the efforts of the degree of damping and the importance of critical damping in cases such as a car suspension system].	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-D-16] Justify qualitatively the factors which determine the frequency response and sharpness of the resonance.	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-D-17] Identify the use of standing waves and resonance in applications [such as Rubens tubes, Chladni plates and acoustic levitation (knowledge of wave harmonic modes is not required)].	Formative	Understanding	Question(s) will not be asked in final examination.	
[SLO: P-12-D-18] Justify the importance of critical damping in a car suspension system.	Summative for theory	Application	Question(s) will be asked in final examination.	
[SLO: P-12-D-19] Justify that there are some circumstances in which resonance is useful [such as tuning a radio, microwave oven and other circumstances in which resonance should be avoided such as airplane's wing or a suspension bridge].	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
Diffraction and Interference:				
[SLO: P-12-D-20] Explain experiments that demonstrate two-source interference using water waves in a ripple tank, sound, light and microwaves.	Summative for theory	Understanding	Question(s) will be asked in final examination.	

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[SLO: P-12-D-21] Describe the conditions required if two-source interference fringes are to be observed.	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-D-22] Use $\Delta y = \frac{\lambda L}{d}$ for double-slit interference using light to solve problems.	Summative for theory	Application	Question(s) will be asked in final examination.	
[SLO: P-12-D-23] Use d sin (θ) = n λ to solve problems.	Summative for theory	understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-D-24] Describe the use of a diffraction grating to determine the wavelength of light [the structure and use of the spectrometer are not included].	Summative for theory	Knowledge +Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-D-25] With the context of the electron diffraction double slit experiment, explain the below two of the many interpretations of quantum mechanics: (i) Copenhagen interpretation (ii) Many worlds interpretation.	Formative	Knowledge + Understanding	Question(s) will not be asked in final examination.	
Domain: E Electricity and Magnetism				
[SLO: P-12-E-01] Define and calculate electric potential [At a point as the work done per unit positive charge in bringing small test charge from infinity to point. Use $V = \frac{q}{4\pi \in_o r}$ for the electric potential in the field due to a point charge].	Summative for theory	Knowledge + Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-E-02] Use the fact that the electric field at a point is equal to the negative of potential gradient at that point.	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-E-03] State how the concept of electric potential leads to the electric potential energy of two-point charges and use $E_p = \frac{Qq}{4\pi \in_o r}$.	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-E-04] Define and calculate capacitance [as applied to both isolated spherical conductors and to parallel plate capacitors].	Summative for theory	Knowledge + Understanding +Application	Question(s) will be asked in final examination.	

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[SLO: P-12-E-05] Derive and apply formulae for the combined capacitance of capacitors in series and in parallel.	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-E-06] Use the capacitance formula for capacitors in series and in parallel.	Summative for theory and PBA	Application	Question(s) will be asked in final theory examination and PBA.	
[SLO: P-12-E-07] Determine the electric potential energy stored in a capacitor from the area under the potential— charge graph [Use $\frac{1}{2}(QV)=\frac{1}{2}(CV^2)$ to solve physics related problems].	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-E-08] Analyze graphs of the variation with time of potential difference, charge and current for a capacitor discharging through a resistor [use τ = RC for the time constant for a capacitor discharging through a resistor].	Summative for PBA	Understanding + Application	Question(s) will be asked in final PBA examination.	
[SLO: P-12-E-09] Use equations of the form $x = x_o\,(\exp(\frac{-t}{RC}))$ [where x could represent current, charge or potential difference for a capacitor discharge through a resistor].	Summative for PBA	Application	Question(s) will be asked in final PBA examination.	
[SLO: P-12-E-10] list the use of capacitors in various household appliances [such as in flash guns, refrigerators, electric fans, rectification circuits, etc.].	Summative for theory	Knowledge	Question(s) will be asked in final examination.	30 periods
Bioelectricity:				
[SLO: P-10-E-11] Illustrate how bioelectricity is generated in animals [- cells control the flow of specific charged elements across the membrane with proteins that sit on the cell surface and create an opening for certain ions to pass through. These proteins are called ion channels. - When a cell is stimulated, it allows positive charges to enter the cell through open ion channels. The inside of the cell then becomes more positively charged, which triggers further electrical currents that can turn into electrical pulses, called action potentials. - The bodies of many organisms use certain patterns of action potentials to initiate the correct movements, thoughts and behaviors].	Formative	Knowledge	Question(s) will not be asked in final examination.	

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[SLO: P-10-E-12] State that there are several species of aquatic life, such as Electrophorus Electricus, that can naturally generate external electric shocks through internal biological mechanisms that act as batteries.	Formative	Knowledge	Question(s) will not be asked in final examination.	
[SLO: P-10-E-13] Explain, with examples of animal with this ability, that electroreception is the ability to detect weak naturally occurring electrostatic fields in the environment.	Formative	Understanding	Question(s) will not be asked in final examination.	
AC circuits:				
[SLO: P-10-E-14] Use the terms period, frequency and peak value as applied to an alternating current or voltage.	Summative for theory	Knowledge + Understanding	Question(s) will be asked in final examination.	
[SLO: P-10-E-15] Use equations of the form $x = x_o \sin(\omega t)$ representing a sinusoidally alternating current or voltage.	Summative for theory	Application	Question(s) will be asked in final examination.	
[SLO: P-10-E-16] Use the fact that the mean power in a resistive load is half the maximum power for a sinusoidal alternating current.	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-10-E-17] Distinguish between root-mean-square (r.m.s.) and peak values [including stating and using $I = \frac{Io}{\sqrt{2}}$ and $V = \frac{Vo}{\sqrt{2}}$ for a sinusoidal alternating current].	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-10-E-18] Distinguish graphically between half-wave and full-wave rectification.	Summative for theory	Knowledge + Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-E-19] Explain the use of a single diode for the half-wave rectification of an alternating current.	Summative for theory and PBA	Understanding	Question(s) will be asked in final theory examination and PBA.	
[SLO: P-12-E-20] Explain the use of four diodes (bridge rectifier) for the full-wave rectification of an alternating current.	Summative for theory and PBA	Understanding	Question(s) will be asked in final theory examination and PBA.	
[SLO: P-12-E-21] Analyze the effect of a single capacitor in smoothing current flow [including the effect of the values of capacitance and the load resistance].	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-E-22] Define mutual inductance (M) and self-inductance (L), and their unit henry.	Summative for theory	Knowledge	Question(s) will be asked in final examination.	

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[SLO: P-12-E-23] Describe the phase of A.C and how phase lags and leads in A.C Circuits.	Summative for theory	Knowledge + Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-E-24] Identify inductors as important components of A.C circuits termed as chokes [devices which present a high resistance to alternating current].	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-E-25] Calculate the reactance of capacitors and inductors.	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-E-26] Describe impedance as vector summation of resistances and reactances.	Summative for theory	Knowledge+ Understanding	Question(s) will be asked in final examination.	
Domain: F Modern Physics				
Quantum Physics:				
[SLO: P-12-F-01] State that electromagnetic radiation has a particulate nature.	Summative for theory	Knowledge	Question(s) will be asked in final examination.	
[SLO: P-12-F-02] Explain and apply the photonic model of light to solve problems [use E = hf to solve problems, and use the electron volt (eV) as a unit of energy].	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-05] Explain that a photon has momentum [including that the momentum is given by p = E/c (connect with the idea that light can exert a force)].	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-06] Describe that photoelectrons may be emitted from a metal surface when it is illuminated by electromagnetic radiation.	Summative for theory and PBA	Understanding	Question(s) will be asked in final theory examination and PBA.	
[SLO: P-12-F-07] Describe and use the terms threshold frequency and threshold wavelength.	Summative for theory and PBA	Understanding + Application	Question(s) will be asked in final theory examination and PBA.	
[SLO: P-12-F-08] Explain photoelectric emission in terms of photon energy and work function energy.	Summative for theory and PBA	Knowledge + Understanding	Question(s) will be asked in final theory examination and PBA.	

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[SLO: P-12-F-09] State and apply $hf = \emptyset + \frac{1}{2}mv^2_{max}$.	Summative for theory and PBA	Understanding +Application	Question(s) will be asked in final theory examination and PBA.	
[SLO: P-12-F-10] Explain why the maximum kinetic energy of photoelectrons is independent of intensity, whereas the photoelectric current is proportional to intensity.	Summative for theory and PBA	Understanding	Question(s) will be asked in final theory examination and PBA.	26 periods
[SLO: P-12-F-11] Juxtapose the evidence for light as a wave and as a particle [Explain that the photoelectric effect provides evidence for a particulate nature of electromagnetic radiation while phenomena such as interference and diffraction provide evidence for wave nature].	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-12] Analyze qualitatively the evidence provided by electron diffraction for the wave nature of particles.	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-13] Explain and apply the de Broglie wavelength to solve problems [use $\lambda = h / p$ to solve problems].	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-14] State that there are discrete electron energy levels in isolated atoms (e.g. atomic hydrogen).	Summative for theory	Knowledge + Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-15] Explain the appearance and formation of emission and absorption line spectra.	Summative for theory	Knowledge + Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-16] Use hf = ΔE to solve problems.	Summative for theory	Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-17] Describe the Compton effect qualitatively.	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-18] Explain the phenomena of pair production and pair annihilation.	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-19] Explain how electron microscopes achieve very high resolution.	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-20] State and explain Heisenberg's uncertainty principle qualitatively.	Summative for theory	Understanding	Question(s) will be asked in final examination.	

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[SLO: P-12-F-21] Use the uncertainty principle to explain why empirical measurements must necessarily have uncertainty in them.	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
Particle Physics:				
[SLO: P-12-F-22] Recognize the equivalence between energy and mass as represented by $E = \Delta mc^2$ and state and use this equation.	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-23] Define and use the terms mass defect and binding energy.	Summative for theory	Knowledge + Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-24] Sketch the variation of binding energy per nucleon with nucleon number.	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-25] Recall what is meant by nuclear fusion and nuclear fission.	Summative for theory	Knowledge	Question(s) will be asked in final examination.	
[SLO: P-12-F-26] Explain the relevance of binding energy per nucleon to nuclear reactions, including nuclear fusion and nuclear fission. [SLO: P-12-F-26] Explain how the neutrons produced in fission create a chain reaction and that this is controlled in a nuclear reactor [including the action of coolant, moderators and control rods].	Summative for theory	Knowledge + Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-27] Calculate the energy released in nuclear reactions using $E = \Delta mc^2$.	Summative for theory	Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-28] Explain that fluctuations in count rate provide evidence for the random nature of radioactive decay.	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-29] Explain that radioactive decay is both spontaneous and random.	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-30] Define activity and decay constant and state the use of $A = \lambda N$.	Summative for theory	Understanding Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-31] Explain half-life with examples.	Summative for theory	Knowledge + Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-32] Use $\lambda = 0.693 / T$ to solve numerical problems.	Summative for theory	Application	Question(s) will be asked in final examination.	

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[SLO: P-12-F-33] State the exponential nature of radioactive decay.	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-34] Use the relationship $x = x_o e^{\lambda t}$ [where x could represent activity, number of undecayed nuclei or received count rate to solve problems analytically and graphically].	Summative for theory	Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-35] Describe the function of the principle components of a water moderated power reactor [core, fuel, rods, moderator, control rods, heat exchange, safety rods and shielding].	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-36] Explain why uranium fuel needs to be enriched before use.	Summative for theory	Knowledge + Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-37] Compare the amount of energy released in a fission reaction with the given energy released in a chemical reaction.	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-38] Explain what is a medical tracer [a substance containing radioactive nuclei that can be introduced into the body and is then absorbed by the tissue being studied].	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-39] Explain annihilation reactions [they occur when a particle interacts with its antiparticle and that mass—energy and momentum are conserved in the process].	Summative for theory	Knowledge	Question(s) will be asked in final examination.	
[SLO: P-12-F-40] Illustrate how PET scanning works [positrons emitted by the decay of the tracer annihilate when they interact with electrons in the tissue, producing a pair of gamma-ray photons traveling in opposite directions].	Formative	Knowledge	Question(s) will not be asked in final examination.	
[SLO: P-12-F-41] Calculate the energy of the gamma ray photons emitted during the annihilation of an electron-positron pair.	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-42] Explain that the gamma-ray photons from an annihilation event travel outside the body and can be detected [including that an image of the tracer concentration in the tissue can be created by processing the arrival times of the gamma-ray photon].	Summative for theory	Understanding	Question(s) will be asked in final examination.	

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[SLO: P-12-F-43] Explain the term luminosity [as the total power of radiation emitted by a star].	Summative for theory	Knowledge+ Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-44] Apply the inverse square law for radiant flux intensity [F in terms of the luminosity L of the source $F = \frac{L}{4\pi d^2}$].	Summative for theory	Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-45] Define and apply standard candles [Explain the use of standard candles to determine distances to galaxies].	Formative	Knowledge + Understanding	Question(s) will not be asked in final examination.	
[SLO:P-12-F-46] Explain blackbody radiation and apply Wien's displacement law to solve problems [$\lambda_{\max} T = constant$ to estimate the peak surface temperature of a star].	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-47] Apply the Stefan Boltzmann law to solve problems [$L = 4\pi r^2 x \sigma T^4$ to solve problems].	Summative for theory	Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-48] Estimate the radius of a star [applying Wien's displacement law and the Stefan—Boltzmann law].	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-49] Explain that the lines in the emission and absorption spectra from distant objects show an increase in wavelength from their known values.	Summative for theory	Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-50] Explain why redshift leads to the idea that the Universe is expanding [include using $\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$ for the redshift of electromagnetic radiation from a source moving relative to an observer to solve problems relating to the expanding universe].	Summative for theory	Understanding +Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-51] State and explain Hubble's law and how it leads to the Big Bang theory.	Formative	Knowledge + Understanding	Question(s) will not be asked in final examination.	
[SLO: P-12-F-52] Describe Earth's climate system as a complex system having five interacting components [the atmosphere (air), the hydrosphere (water), the cryosphere (ice and permafrost), the lithosphere (earth's upper rock layer) and the biosphere (living things)].	Summative for theory	Knowledge + Understanding	Question(s) will be asked in final examination.	

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[SLO: P-12-F-53] Relate ocean currents and wind patterns to the climate system [as the statistical characterization of the climate system, representing the average weather, typically over a period of 30 years, and is determined by a combination of processes in the climate system, such as ocean currents and wind patterns].	Summative for theory	Knowledge + Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-54] Explain climate inertia [as the phenomenon by which climate systems show resistance or slowness to changes in significant factors e.g. stabilization of greenhouse emissions might show a slow response due to the action of complex feedback systems].	Summative for theory	Knowledge + Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-55] Explain that climate change can be categorized into internal variations and external forcing.	Summative for theory	Knowledge + Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-56] Explain how global climate is determined by energy transfer from the Sun [with specific reference to the below factors and terms: state and use the term Earth energy budget. Explain how the energy imbalance between the poles and the equator can affect atmospheric circulation].	Summative for theory	Knowledge + Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-57] Explain that due to the conservation of angular momentum, the Earth's rotation diverts the air to the right in the Northern Hemisphere and to the left in the Southern hemisphere, thus forming distinct atmospheric cells.	Summative for theory	Knowledge + Understanding	Question(s) will be asked in final examination.	18 periods
[SLO: P-12-F-58] Explain that ocean water that has more salt has a higher density and differences in density play an important role in ocean circulation.	Summative for theory	Knowledge + Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-59] Explain how the thermohaline circulation transports heat from the tropics to the polar regions.	Formative	Knowledge	Question(s) will not be asked in final examination.	
[SLO: P-12-F-60] Explain how climate science is an example of a chaotic system, [using the metaphor of a butterfly" wing flaps may cause hurricanes in another part of the world, mathematics of chaos theory is not required; just the idea that with very complex systems it is very difficult to predict outcomes and they are very sensitive to initial conditions].	Formative	Knowledge + Understanding	Question(s) will not be asked in final examination.	

NCP SLOs Description	Form of Assessment	Cognitive domain	Remarks	Number of Periods Required (1 period=40 minutes)
[SLO: P-12-F-61] Explain that piezo-electric effect and its application in medical science [ultrasound waves are generated and detected by a piezoelectric transducer].	summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-62] Explain how ultrasound can be used to obtain diagnostic information about internal body structures.	summative for theory	Knowledge + Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-63] Explain that X-rays are produced by electron bombardment of a metal target and calculate the minimum wavelength of X-rays produced from the accelerating P.d.	Summative for theory	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-12-F-64] Explain the use of X-rays in imaging internal body structures [including a describing of the term contrast in X-ray imaging].	Summative for theory	Knowledge +Understanding	Question(s) will be asked in final examination.	
[SLO: P-12-F-65] Explain how computed tomography (CT) scanning works [it produces a 3D image of an internal structure by first combining multiple X-ray images taken in the same section from different angles to obtain a 2D image of the section, then repeating this process along an axis and combining 2D images of multiple sections].	Formative	Knowledge	Question(s) will not be asked in final examination.	
Domain: G Nature of Sciences				
Debates about Beauty in Physics:				
[SLO: P-12-G-01] Explain, with examples, what do thinkers who hold the view that there is inherent mathematical beauty in the natural world mean by: (i) elegance of simplicity (ii) symmetry.	Formative	Knowledge	Question(s) will not be asked in final examination.	
[SLO: P-12-G-02] Explain, with an example, a counterargument to the claim that physical truths must be inherently mathematically elegant or display symmetry.	Formative	Application	Question(s) will not be asked in final examination.	
[SLO: P-12-G-03] Describe the main pros and cons in the debate about: (i) Whether humans should research whether there are aliens somewhere in the universe. (ii) Whether research should continue on uncovering the secrets of	Formative	Knowledge + Understanding	Question(s) will not be asked in final examination.	

NCP SLOs Description	Form of Assessment	Cognitive domain	Remarks	Number of Periods Required (1 period=40 minutes)
subatomic particles, given the advent of nuclear weapons.				
[SLO: P-11-G-04] Explain how the below thought experiments helped convey important physics concepts that would have been impractical to investigate empirically: (i) Newton's cannonball.	Formative	Understanding	Question(s) will not be asked in final examination.	
Domain: H Experimentation Skills				
[SLO: P-12-N-01] Develop and justify safety guidelines for a proposed procedure, that also outline the overall risks of the experiment, keeping in mind: [(i) the apparatus, (ii) the surrounding environment, (iii) need for personal protective equipment].	Formative for PBA	Understanding + Application	Question will not be asked in final examination, however, it will be part of Lab work.	
[SLO: P-12-N-02] Formulate a testable hypothesis by: « Identifying the independent variable in the experiment « Identifying the dependent variable in the experiment « Identifying the variables that are to be kept constant.	Formative for PBA	Understanding	Question will not be asked in final examination, however, it will be part of Lab work.	08 periods
[SLO: P-12-N-03] Explain the methods of data collection by: a. describing the method to be used to vary the independent variable b. describing how the independent and dependent variables are to be measured c. describing how other variables are to be kept constant d. describing, with the aid of a clear labeled diagram, the arrangement of apparatus for the experiment and the procedures to be followed.	Formative for PBA	Understanding	Question will not be asked in final examination, however, it will be part of Lab work.	
[SLO: P-12-N-04] Explain the methods of data analysis by describing how the data should be used in order to reach a conclusion, including details of derived quantities to be calculated from graphs.	Formative for PBA	Understanding	Question will not be asked in final examination, however, it will be part of Lab work.	